



## Listing factors

### TASK A : Systematic approach to find factor pairs

1) List the factor pairs of these numbers.

a) 18

$$1 \times 18$$

$$2 \times 9$$

$$3 \times 6$$

b) 28

$$1 \times 28$$

$$2 \times 14$$

$$4 \times 7$$

c) 36

$$1 \times 36$$

$$2 \times 18$$

$$3 \times 12$$

$$4 \times 9$$

d) 32

$$1 \times 32$$

$$2 \times 16$$

$$4 \times 8$$

e) 100

$$1 \times 100$$

$$2 \times 50$$

$$4 \times 25$$

$$5 \times 20$$

f) 67

$$1 \times 67$$

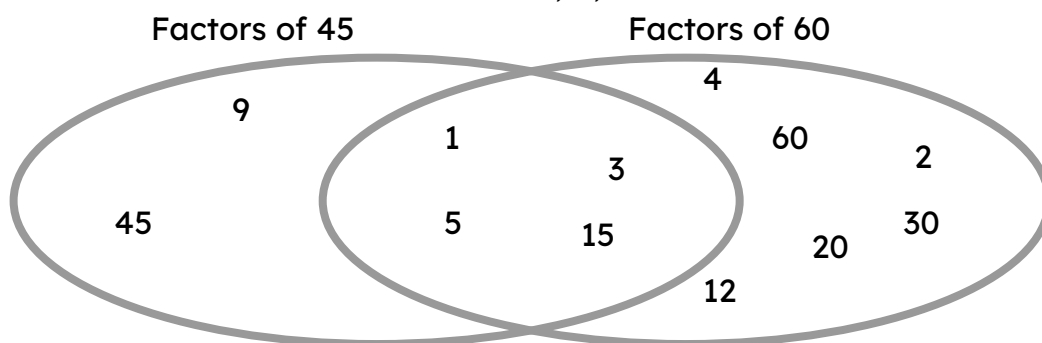
2) Sam makes two rectangles with an area of  $10 \text{ cm}^2$ .  
Explain how these rectangles show all of the factors of 10.



*Because you cannot make a rectangle with a height of 3, 4, 6, 7, 8, or 9.  
So 1 & 10 and 2 & 5 are the only factor pairs.*

## TASK B : Common factors

- 1) Use the Venn diagram to fill in the gaps.
  - a) 45 has 2 factors which are not common with 60.
  - b) 60 has 6 factors which are not common with 45.
  - c) 45 and 60 have 4 common factors.
  - d) The common factors of 45 and 60 are 1, 3, 5 and 15.



- 2) Find the common factors for each of these following pairs of numbers:

a) 24 & 36

$$\begin{array}{l} 1 \times 24 \\ 2 \times 12 \\ 3 \times 8 \\ 4 \times 6 \\ 1 \times 36 \\ 2 \times 18 \\ 3 \times 12 \\ 4 \times 9 \\ 6 \times 6 \end{array}$$

b) 24 & 38

$$\begin{array}{l} 1 \times 24 \\ 2 \times 12 \\ 3 \times 8 \\ 4 \times 6 \\ 1 \times 38 \\ 2 \times 19 \end{array}$$

c) 24 & 40

$$\begin{array}{l} 1 \times 24 \\ 2 \times 12 \\ 3 \times 8 \\ 4 \times 6 \\ 1 \times 40 \\ 2 \times 20 \\ 4 \times 10 \\ 5 \times 8 \end{array}$$

d) 24 & 72

$$\begin{array}{l} 1 \times 24 \\ 2 \times 12 \\ 3 \times 8 \\ 4 \times 6 \\ 1 \times 72 \\ 2 \times 36 \\ 3 \times 24 \\ 4 \times 18 \\ 6 \times 12 \\ 8 \times 9 \end{array}$$

## TASK C : Simplifying fractions

- 1) Simplify fully the following fractions.

Try to use the highest common denominator.

a)  $\frac{5}{45}$

$$\frac{5 \times 1}{5 \times 9} = \frac{1}{9}$$

b)  $\frac{12}{15}$

$$\frac{3 \times 4}{3 \times 5} = \frac{4}{5}$$

c)  $\frac{30}{36}$

$$\frac{6 \times 5}{6 \times 6} = \frac{5}{6}$$

d)  $\frac{40}{72}$

$$\frac{8 \times 5}{8 \times 9} = \frac{5}{9}$$

e)  $\frac{36}{54}$

$$\frac{18 \times 2}{18 \times 3} = \frac{2}{3}$$

- 2) a) Which of the following are not in their simplest form?  
b) Simplify those you have identified.

$$\frac{14}{35} \quad \frac{7 \times 2}{7 \times 5} = \frac{2}{5} \quad \frac{3}{11}$$

$$\frac{9}{16}$$

$$\frac{2}{19}$$

$$\frac{12}{33} \quad \frac{3 \times 4}{3 \times 11} = \frac{4}{11}$$